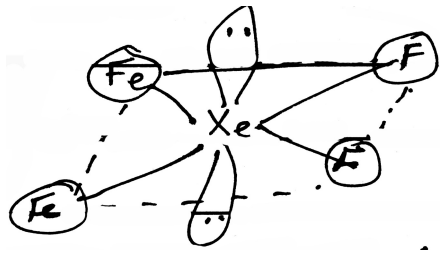
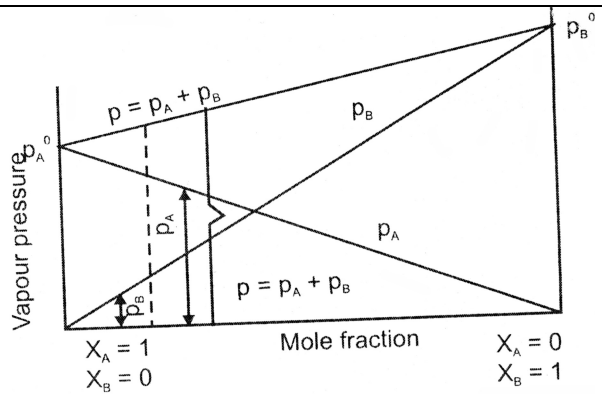
CODE: 67/OSS

MARKING SCHEME
SENIOR SECONDARY
MARCH-APRIL 2024
CHEMISTRY
(313)

SET			EXPECTED VALUE POINTS FOR EACH STEP	Distribution Of Marks	Total Marks
A	B	C			
Q. No.					
			SECTION A		
1	-	-	A	1	1
2	-	-	(i) C OR (ii) C	1	1
3	-	-	A	1	1
4	-	-	(i) C OR (ii) B	1	1
5	-	-	B	1	1
6	-	-	(i) B OR (ii) D	1	1
7	-	-	C	1	1
8	-	-	(i) B OR (ii) D	1	1
9	-	-	D	1	1
10	-	-	(i)A OR(ii) B	1	1
11	-	-	C	1	1
12	-	-	(i) C OR (ii) B	1	1
13	-	-	(i) C OR (ii) C	1	1
14	-	-	(i) A OR (ii) A	1	1
15	-	-	C	1	1
16	-	-	A	1	1
17	-	-	(1) True (2) False (3) True (4) True	1x2	2
18	-	-	(1) p- orbitals l = 1 (2) m_l = -3, -2, -1, 0, +1, +2, +3 (3) Two (4) Wave function describes the motion of an electron inside on atom.	1x2	2

19	-	-	(1) Bonding and anti-bonding M.O. (2) Aufban principle – The electrons are filled in the increasing order of their energies in orbitals. (3) V.B.T. visualizes the bond formation to be localized whereas according to M.O.T. it is delocalized. (4) Bond order = ½ (2-2) = 0 for Be molecule, hence it does not exist.	1x2	2
20	-	-	(1) At low concentrations, the magnitude of osmotic pressure is large enough for measurement. (2) Solutions having equal osmotic pressure. Example: 0.1 m glucose and 0.01 m sucrose solution are isotonic. (Or any other example) (3) Due to osmosis through the roots because root cell walls are made up of semi permeable membrane. (4) Due to osmosis, water molecules move into blood cells through the cell wall and they swell.	1/2×2	2
21	-	-	B	2	2
22	-	-	(1) Group 1 and Group 2 metals CaH_2 / NaH / or any other. (2) $ZnO + H_2 \rightarrow Zn + H_2O$ (or any other example) (3) These are compounds formed by a number of transition metals in which hydrogen is accommodated in the lattice of the transition element. (4) Because it could replace gasoline to power automobiles / be used in the oxygen gas in fuel cells to generate electricity / it has maximum calorific value.	1x2	2

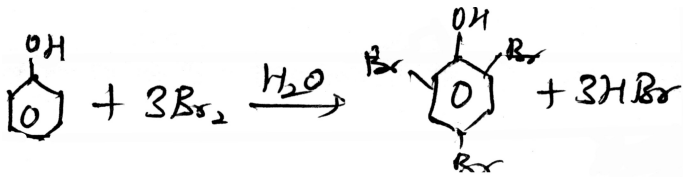
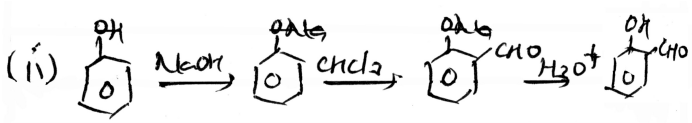
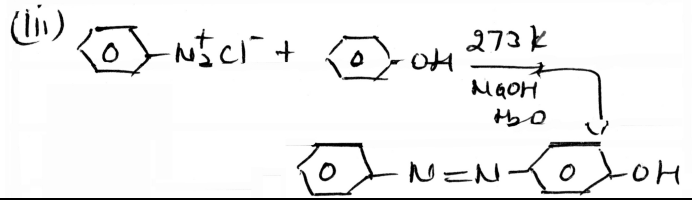
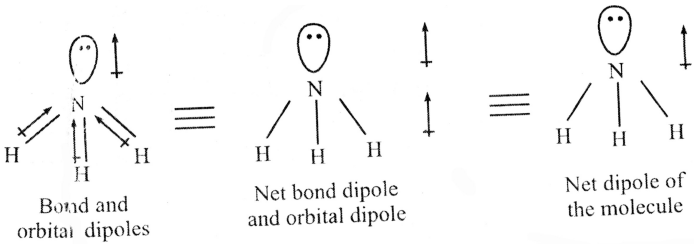
23	-	-	(1) increases (2) metallic (3) +1 (4) fluorine	1x2	2
24	-	-	(1) Compound of Xenon with PtF_6. (2) Helium, Neon, Argon (3) $\underset{2}{Xe} + \underset{1}{F_2} \xrightarrow[1bar]{873k} XeF_2$ (4) 	1x2	2
25	-	-	C a – (iii); b (i); c (iv); d-(ii)	2	2
26	-	-	1. dry ether 2. free radical 3. alc-KOH 4. Calcium Carbide	1x2	2
27	-	-	A ***(Note: Give marks to all the learners who attempt this question)***	1x2	2
28	-	-	(1) Glycogen (2) 10 (3) derived (4) Pentose	1x2	2
			SECTION B		
29	-	-	Ideal solution – A solution which obeys Rasult's low over the entire range of concentration at all temperatures is called an ideal solution.	1	2

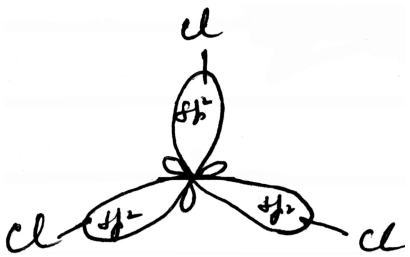
			 OR Some of the molecules of the solvent on the surface are replaced by the molecules of the solute which are non-volatile. The vapour phase of such a solution consists of vapours of the solvent in solution. Since the mole fraction of vapours in the solution is less than one, therefore the vapour pressure of the solution will be less than that of pure solvent.	1 2	
30	-	-	(i) $\Delta_r H = [2\Delta_f H°(HCl) + \Delta_f H°(f_2)] -$ $[2\Delta_f H°(Hf) + \Delta fH°(Cl_2)]$ $= [2\times(-92.5)+0]-[2\times(-269)+0]kJ$ $= -185kJ + 538kJ$ $= +353kJ$ OR (ii) (a) Heat evolved when one mole of an acid reacts with one mole of a base to form salt and water. (b) Because a strong acid and a strong base dissociates completely in solution and their neutralization enthalpy is always constant.	1 ½ ½ 1 1	2
31	-	-	An acid is a substance that can donate a	½+ ½	2

			proton whereas a base is a substance that can accept a portion Example : $M_2O + NH_3 \square NH_4^+(aq) + OH(aq)$ $acid_2$ $base_2$ $acid_2$ $base_2$ (or any other example)	1	
32	-	-	Factors: (1) The increase in the promotion energy from the ground state (ns^2 np^1) to the valance state (ns^1 np^2). (2) Power overlap of the orbitals of the large atoms and hence power energy. OR $PbCl_4$ is covalent. The covalent character of a halide increases with the increase in the oxidation state of the element. $PbCl_4$, Pb = + 4 $PbCl_2$ Pb = + 2 It is prepared by the oxidation of chlorides by strong oxidizing agents such as MnO_2.	1x2 ½ + ½ ½ ½	2
33	-	-	$MnO_2 + 2Cl^- + 4M^+ \rightarrow Mn^{2+} + 2H_2O + Cl_2$ (or any other equation) Use: Used in bleaching industry / manufacture of plastics, synthetic rubbers, antiseptic and insecticides Or any other Any two	1 ½ + ½	2
34	-	-	(1) 2,3 Dimethyl but -1-ene	1	2

			(2) 1-Ethyl-3-methylbenzene	1	
35	-	-	(i) $R-\overset{O}{\overset{\|\|}{C}}-OH + R'NH_2 \rightarrow R-\overset{O}{\overset{\|\|}{C}}-NHR' + H_2O$ $CH_3COCl + CH_3NH_2 \rightarrow CH_3-\overset{O}{\overset{\|\|}{C}}-NH-CH_3$ Or any other **OR** (ii) Wolff-Kishner reduction - $C_6H_5COCH_3 \xrightarrow[2.\ H_2O]{1.\ NH_2NH_2, -OH, \Delta} C_6H_5CH_2CH_3$ Clemmenton reduction – $CH_3(CH_2)_5CHO \xrightarrow[Hcl, H_2O)]{Zn(Hg)} CH_3(CH_2)_5-CH_3$ Heptanal Heptane (or any other)	1 1 1 1	2
36	-	-	The reaction should be carried out using excess of ammonia. $C_2H_5Br + 2NH_3 \rightarrow C_2H_5-NH_2 + NH_4Br$ (Or any other equation)	1 1	2
37	-	-	Vulcanization: Addition of sulphur to hot rubber causes changes that improves its physical properties. It is carried out by heating crude rubber in the prescribing sulphur.	1 1	2
38	-	-	15 g of H_2SO_4 = 0.153 moles (98 g of H_2SO_4 = 1 mole) Volume of Solution = $\frac{\text{Mass of solution}}{\text{Density of solution}}$ $= \frac{100}{1.02} = 98.04\,cm^3$ $= \frac{98.04}{1000} L = 0.09804$	½ ½ ½ ½	3

			Molality (M) = $= \frac{nB}{V\ (d)} = \frac{\text{no of moles of solute}}{\text{volume of solution}}$ $= \frac{0.153\,mol}{0.09804\,L} = 1.56\,mol\ L^{-1}$ $= 1.56M$ **OR** Solute ($C_{12}H_{22}O_{11}$) = 34.2g Solution = 214.2g Solvent = 214.2 – 34.2 = 180g = 0.180 kg (i) Molal conc $= \frac{34.2/342\ mol}{0.180\,kg} = 0.556\,mol\,kg^{-1}$ (ii) Mole fraction $= \frac{34.2/342}{\frac{180}{18} + \frac{34.2}{342}} = 0.0099$	½ ½ 1 1 1	
39	-	-	$MnO_2 + 2KOH \rightarrow K_2MnO_3 + H_2O$ Pyrolusite $2K_2MnO_3 + O_2 \rightarrow 2K_2MnO_4$ $3MnO_4^{2-} + 4H+ \rightarrow 2MnO4^- + MnO_2 + 2H_2O$ or $2K_2MnO_4 + Cl_2 \rightarrow 2kMnO_4 + 2KCl$	1 1 1	3
40	-	-	In $[Ni(H_2O)_6]^{2+}$, Ni is in +2 state with the configuration $3d^s$ i.e. it has two unpaired C^- which do not pair up in the presence of the weak H_2O ligand. The d-d transition absorbs red light and the complementary light emitted is green. In $[Ni\ (CN)_4]^{2-}$, $3d^8$ i.e. it has two unpaired c^- which do not pair up in the presence of the weak H_2O ligand. The d-d transition absorbs red light and the complementary light emitted is green. In $[Ni(CN)_4]^{2-}$, $3d^8$ but CN^- is a strong ligand and cause pairing of electrons. Thus no	½ ½ ½ ½ ½ ½	3

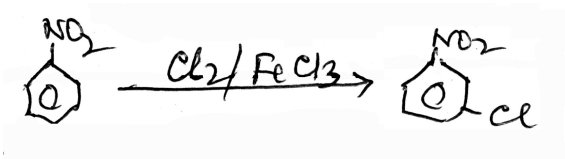
			unpaired e^- Hence colourless.		
41	-	-	(i)   	1 1 1	3
42	-	-	(i) (1) In a covalent bond the shared pair of electrons is closer to the mere electro negative atom. This leads to change separation in the molecule and the bond becomes polar. Example $\overset{\delta+}{H}-\overset{\delta-}{Cl}$ (2) The three N-H bonds in ammonia are unsymmetrical. It has a pyramidal structure. In each N-H bond, nitrogen is the negative centre and hydrogen is positive centre. In addition to it N has a lone pair of electrons which also has a orbital dipole moment. Figure on Page 106 Book I (iv)  OR	1 1 3	5

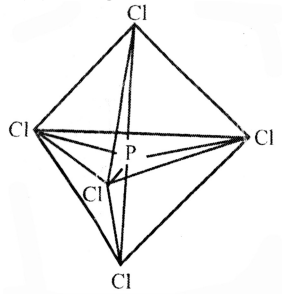
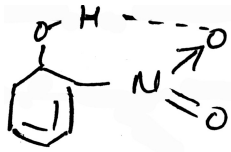
			(ii) (1) Hybridization is the process of forming new orbitals by mixing of atomic orbitals in a particular atom. Characteristics: (1) The number of hybrid orbitals formed is the same as the number of atomic orbitals undergoing hybridization. (2) All the new hybrid orbitals that are formed are exactly identical in their shapes and energy. (2) B = 5 = $1s^2 2s^2 2p^1$ (ground state) = $1s^2\ 2s^1\ 2p_2^1\ 2p_y^1\ 2p_2$ (excited state) 1 sp^2 hybridization  equilateral triangle The three sp^2 hybrid orbitals form bonds with the p-orbitals of chlorine atoms and the shape is planar.	1 ½ ½ ½ ½ 1 1	
43	-	-	(i) (1) Kohlronachiv law – At infinite dilution, each ion of the electrolyte makes a definite contribution towards conductivity of the electrolyte and it is independent of the presence of other ions of the electrolyte. (2) $\wedge^{\circ}_{m}$ for $Ba(OH)_2 = \lambda^{\circ}_{Ba^{2+}} + 2\lambda^{\circ}_{OH^-}$ ...(i) $\wedge^{\circ}_{m}$ for $BaCl_2 = \lambda^{\circ}_{Ba^{2+}} + 2\lambda^{\circ}_{Cl^-}$ .... (ii) $\wedge^{\circ}_{m}$ $(NH_4Cl) = \lambda^{\circ}_{MH_4^+} + \lambda^{\circ}_{Cl^-}$ ... (iii)	2 1 ½ ½ 1	5

			$\wedge^{\circ}_{m}(NH_4OH) = \lambda^{\circ}_{MH_4^+} + \lambda^{\circ}_{OH^-}$ $= \frac{1}{2}\text{eqn.}(1) + \text{eqn.}(iii) - \frac{1}{2}\text{ eqn. (ii)}$ $= \frac{1}{2} \times 457.6 + 129.8 - \frac{1}{2} \times 240.6$ $= 238.3\,\Omega^{-1}\,cm^2\,mol^{-1}$ OR (ii) (1) It is a series in which the standard potential of a large number of electrodes are listed in the increasing order of electrode potential. (2) The difference in the potential of the two electrodes of a galvanic cell when measured in an open circuit is cell emf. When it is measured in a closed circuit with some external load it is called potential difference. (3) $E^{\circ}_{Cell} - E^{\circ}_{Cu^{2+}/cu} - E^{\circ}_{Cu^{2+}/Ni}$ $0.59 = 0.34 - E^{\circ}_{Ni^{2+}/Ni}$ or $E^{\circ}_{Ni^{2+}/Ni} = 0.34 - 0.059$ $= -0.25\text{volt}$	1 1 1 2	

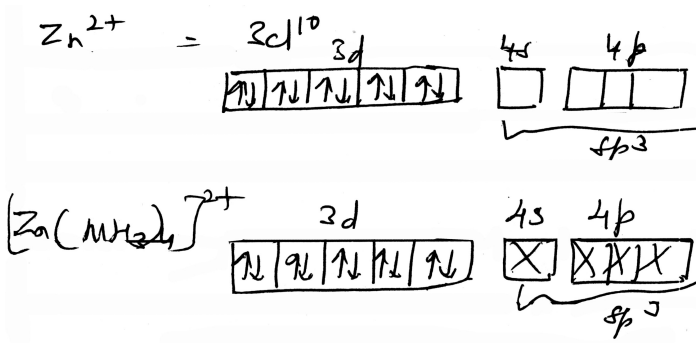
			SET B		
			SECTION A		
-	1	-	B	1	1
-	2	-	Same as Q. 8 of A1	1	1
-	3	-	Same as Q. 6 of A1	1	1
-	4	-	Same as Q. 7 of A1	1	1
-	5	-	Same as Q. 9 of A1	1	1
-	6	-	Same as Q. 16 of A1	1	1
-	7	-	Same as Q. 15 of A1	1	1
-	8	-	Same as Q. 2 of A1	1	1
-	9	-	Same as Q. 4 of A1	1	1
-	10	-	Same as Q. 3 of A1	1	1
-	11	-	Same as Q. 5 of A1	1	1
-	12	-	Same as Q. 14 of A1	1	1
-	13	-	Same as Q. 10 of A1	1	1
-	14	-	Same as Q. 11 of A1	1	1
-	15	-	Same as Q. 13 of A1	1	1
-	16	-	Same as Q. 12 of A1	1	1
-	17	-	Same as Q. 20 of A1	2	2
-	18	-	Same as Q. 22 of A1	2	2
-	19	-	Same as Q. 27 of A1	2	2
-	20	-	Same as Q. 28 of A1	2	2
-	21	-	Same as Q. 17 of A1	2	2
-	22	-	Same as Q. 18 of A1	2	2
-	23	-	Same as Q. 26 of A1	2	2
-	24	-	Same as Q. 21 of A1	2	2
-	25	-	Same as Q. 19 of A1	2	2
-	26	-	Same as Q. 23 of A1	2	2
-	27	-	Same as Q. 24 of A1	2	2
-	28	-	Same as Q. 25 of A1	2	2
			SECTION B		
-	29	-	Same as Q. 32 of A1	2	2
-	30	-	Same as Q. 35 of A1	2	2

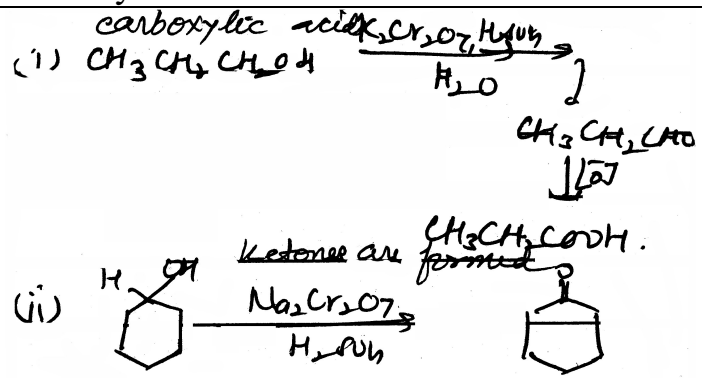
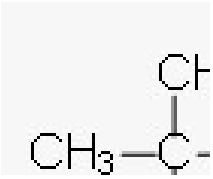
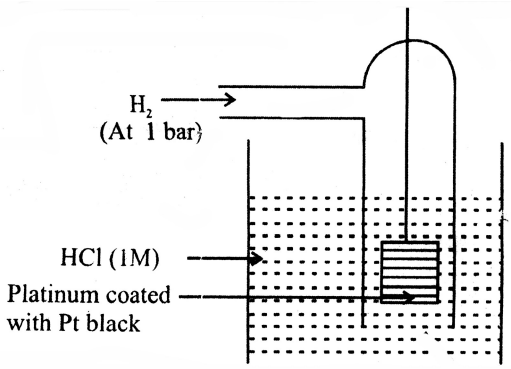
-	31	-	A weak acid $NA_{(aq)} + H_2O_{(l)} \square \quad H_3O^+_{(aq)} + A^-_{(aq)}$ Inital C 55 0 0 (moles) $eq^m_{conc.}$ $C(1-d)$ 55 $C\alpha$ $C\alpha$ $k_a = \frac{[N_3O^+][A^-]}{[H_2O][HA]} = \frac{C\alpha \times C\alpha}{C(1-\alpha)\times 55}$ Rearranging we get: $55 = ka = \frac{[C\alpha][C\alpha]}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$ Since the acid is a weak acid $2 << 1$, We can neglect α in comparison to $\therefore ka = C\alpha^2$ or $\alpha^2 = \frac{ka}{C}$ or $\alpha = \sqrt{\frac{ka}{C}}$	½ ½ ½ ½	2
-	32	-	Same as Q. 29 of A1	2	2
-	33	-	(a) It is prepared by reducing hot conc. H_2SO_4 with Cu turnings. $Cu + 2H_2SO_4 \rightarrow CuSO_4 + 2H_2O + SO_2$ (Or any other method) (b) It oxidizes H_2S to S $2H_2S + SO_2 \rightarrow 2H_2O + 3S$ (Or any other) $Na + X \rightarrow Na \times$ (X=Cl, Br I) By focusing the compound with sodium metal which converts the element present in the compound to ionic firm.	1 ½ +½	2
-	34	-	L.E. is acidized with $HaNO_3$ and then treated with $AgNO_3$. A pale yellow precipitate partially soluble in NH_4OH indicates bovanine.	1	2

			$AgNO_3 + NaBr \rightarrow AgBr + NaNO_3$ (yellow ppr)	1	
-	35	-	Same as Q. 30 of A1	2	2
-	36	-	(i) $CH_3-CH_2-\overset{\overset{O}{\|\|}}{C}-NH_2 \underset{\downarrow}{+} Br_2 + KOH$ $CH_3CH_2NH_2 + 2KBr + K_2CO_3 + 2H_2O$ (ii) 	1 1	2
-	37	-	Same as Q. 37 of A1	2	2
-	38	-	Same as Q. 38 of A1	3	3
-	39	-	Prepn of $K_2Cr_2O_7$ – $4FeO.Cr_2O_3 + 8Na_2CO_3 + 7O_2 \rightarrow 2Fe_2O_3 + 8Na_2CrO_4 + 8CO_2$ $2Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$ $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$	 1 1 1	3
-	40	-	F^- is a weak field ligand therefore, electron will be filled $t_{2g}^4 e_g^{02}$ (four unpaired e^-) $\therefore$ paramagnetic CN^- is a strong field ligand $\therefore$ $t_{2g}^6 e_g^0$ (no unpaired e^-) $\therefore$ diamagnetic	½ 1 ½ 1	3
-	41	-	Same as Q. 41 of A1		
-	42	-	(i) P = 15 = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ ……. $= 1s^2\ 2s^2\ 2p^6\ 3s^1\ 3p_x^1\ 3p_y^1\ 3d_{xy}^1$ (excited taste) $sp^3d\ sp^3d\ sp^3d\ sp^3d\ sp^3d$ Therefore directed towards the corners of a trigonal bipyromide, overlap with 5 chlorine atoms and five σ bonds are formed.	½ ½ 1 1	5

			∴ PC/s has trigonal bipyromidal geometry. Shape Fig 4.5 (a) Page 119 Book 1.  OR (ii) (1) C – C < e = c < C ≡ C (2) Hydrogen bonding is the electrostatic interaction between the positively charged hydrogen atom and the negatively charged electronegative atom. ∴ Intermolecular, intramolecular Intermolecular H – F ---- H – F ---- H – F – H – F--- Intramolecular  (or any other suitable example)	1 1 1 1 ½, ½ 1 1	
-	43	-	Same as Q. 43 of A1	5	5

			SET C		
			SECTION A		
-	-	1	Same as Q. 10 of A1	1	1
-	-	2	Same as Q. 11 of A1	1	1
-	-	3	Same as Q. 16 of A1	1	1
-	-	4	Same as Q. 15 of A1	1	1
-	-	5	C	1	1
-	-	6	Same as Q. 9 of A1	1	1
-	-	7	Same as Q. 12 of A1	1	1
-	-	8	Same as Q. 13 of A1	1	1
-	-	9	Same as Q. 14 of A1	1	1
-	-	10	Same as Q. 1 of A1	1	1
-	-	11	Same as Q. 8 of A1	1	1
-	-	12	Same as Q. 7 of A1	1	1
-	-	13	Same as Q. 2 of A1	1	1
-	-	14	Same as Q. 6 of A1	1	1
-	-	15	Same as Q. 3 of A1	1	1
-	-	16	Same as Q. 4 of A1	1	1
-	-	17	Same as Q. 20 of A1	2	2
-	-	18	Same as Q. 22 of A1	2	2
-	-	19	Same as Q. 28 of A1	2	2
-	-	20	Same as Q. 23 of A1	2	2
-	-	21	Same as Q. 24 of A1	2	2
-	-	22	Same as Q. 25 of A1	2	2
-	-	23	Same as Q. 26 of A1	2	2
-	-	24	Same as Q. 27 of A1	2	2
-	-	25	Same as Q. 21 of A1	2	2
-	-	26	Same as Q. 18 of A1	2	2
-	-	27	Same as Q. 17 of A1	2	2
-	-	28	Same as Q. 19 of A1	2	2

			SECTION B		
-	-	29	Same as Q. 32 of A1	2	2
-	-	30	Same as Q. 29 of A1	2	2
-	-	31	Water con. act as a very weak acid and also a very weak base. $H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$ $K = \frac{[H_3O^+][OH^-]}{[H_2O]^2}$ $[H_3O^+][OH^-] = K_{eq} \times [H_2O)^2]$ $= K_w$ ionic product constant of water	½ ½ ½ ½	2
-	-	32	Same as Q. 30 of A1		
-	-	33	(1) Rhombic and monoclinic sulphur Temperature = 95.6°C (2) $2\ KMnO_4 + 5SO_2 + 2H_2O \rightarrow K_2SO_4 + 2MnSO4 + H2SO4$	½ ½ 1	2
-	-	34	(i) Potential (ii) Geometrical	1 1	2
-	-	35	Same as Q. 37 of A1	2	2
-	-	36	Same as Q. 35 of A1	2	2
-	-	37	Same as Q. 36 of A1	2	2
-	-	38	Same as Q. 39 of A1	3	3
-	-	39	Same as Q. 38 of A1	3	3
-	-	40	$Zn = 30 = 3d^{10}\ 4s^2$ $Zn^{2+} = 3d^{10}$  Geometry – Tetrahedral Magnetic character – Diomagnetic Tetra-ammine zinc (II) ion	½ ½ ½ ½ 1	3

			Carboxylec acid all formed		
-	-	41	 OR (i) Alcohol react with helium or potassium metals to give the respective alkoxides. $CH_3CH_2OH + 2Na \rightarrow CH_3CH_2ONa + \frac{1}{2}H_2$ (ii) 2-chloro-2-methylpropane/structure 	½ 1 ½ 1	3
-	-	42	Same as Q. 42 of A1	5	5
-	-	43	(a) When the concentration of the electrolyte is one molar and the temperature is 298K, then the electrode potential is standard. (b) SHt – consists  A container, containing 1 M HCl at 298 K. A wire containing platinum electrode	1 1 1½	5

			coated with Pt black is immersed in the solution pure H_2 gas is bubbled in the solution at 1 bar pressure / or labeled diagram. Working : E^o is taken as zero it may act as cathode or anode depending on the nature of the other electrode. OR (a) Because it is consumed chemically when current is drawn from it. (b) Zinc cup – Anode MnO_2 + Carbon black + NH_4Cl paste – Electrolyte Carbon rod-cathode. Oxidation - $Zn \rightarrow Zn^{2+} + 2e^-$ Reduction – $2MnO_2 + NH4_4^+ + e^- \rightarrow NnO(OH) + NH_3$ H ydrogen – Oxygen fuel cell	1½ 1 1 1 1 1	